BRETT KNOX

Homework 4

DECEMBER 1, 2025

Stack : Python (pycharm) &SQL (postgres)

Challenges :many , but to name a few – had a lot of trouble actually connecting to the database since it would say it was accessing public but not the eccomerce stuff , once I did access it I was having issues inserting things in since the ID's weren't being generated so I was getting nulls in a non-nullable value, the actual code just wouldn't work since I had a syntax issue that took me a frustrating amount of time to catch as well as some errors from forgetfulness (forgot to add the functions to SQL and it wasn't running as I wanted and was confused for a while) , and more minimal stuff along the way.

Schema:

Categories (caregoryID PK, categoryName, description, isActive)

Products(productID PK, productName, categoryID FK, price, stockQuantity, description, brand, weight, isActive),

Customers(customerID PK, firstName, lastName, email, phone, dateOfBirth, registrationDate, isActive) ,

Orders(orderID PK, CustomerID FK, orderDate, totalAmount, orderStatus, shippingAddress, paymentMethod),

OrderItems(orderID FK, productID FK, quantity, unitPrice, subtotal, (orderID , productID) PK)

Reviews(reviewID PK, customerID FK, productID FK, rating, reviewText, reviewDate, isVerified)

Approach :

I added onto the eccomerce SQl from past assignments & made it so that when products, customers, reviews ,or orders are inserted they have a id automatically generated. I also used the python outline provided and filled in the methods inside. I did this instead of having the python query for the max id for each and create one using max(ID) + 1 since I found this method less error prone (I tried the other way and couldn't get it to work as desired) .

I did not do a gui or any of the extra credit items as I was having a lot of issues with getting things to insert & connect, which I resolve at the 11th hour (6pm on Wednesday) and doubted I could get the rest done in time – so there is a unfilled analytics portion where I was going to do the visualization parts.

I added some triggers and functions to sql (the functions were essential & were described in the outline).

DatabaseConnection – connects you to the database – connects to postgres , and uses queries to prevent injection

ProductManagement – does all the product related options available in the menu (has all the sql queries and functions you call when you select the option)

CustomerManagement – does all the customer related operations , similar to the one above but with the customers table values

OrderManagemenr – does all order related operations , similar to the one above but with the orders table

ReviewManagement – does all review related operations , similar to the one above but with the reviews table

StoredProcedures – does the python wrappers around the SQL stored procedures & functions required by the assignment (PlaceOrder, calculateOrderTotal, checkProductStock, getRestockAlerts)

All the functions that read / write to the table have validation to prevent issues / crashing

Design decisions:

I did a text based CLI instead of a GUI

I did a soft delete for products and customers to keep the referential integrity for past orders and reviews . They are still hidden from the normal searches but not really gone .

The application does validation checks before inserting an item even if the stored function is not used directly.

Implementation steps:

1. Run the SQL that creates the database and the 6 tables.
2. Run the code that fills in the tables (if you want something to test whether the retrieval based options are functional
3. Run the triggers and functions so the python code will run whenever you run it
4. Run the set ecommerce as public & the alter tables lines in SQL so that the python will actually connect to the database & that any new insert will have a unique ID
5. Edit the naming in the DatabaseConnection section in the python

6. Run the python & do any of the functions in the menu by inserting the corresponding number (though option 5 will not work since I didn't fill that part in (non-essential anyway)) .

   Output to show it runs :
   C:\Users\brett\PycharmProjects\EcommerceApp\venv\Scripts\python.exe
   C:\Users\brett\PycharmProjects\EcommerceApp\main.py
   Starting E-Commerce Application...
   Database connection successful!
   E-COMMERCE MANAGEMENT SYSTEM
   1. Product Management
   2. Customer Management
   3. Order Processing
   4. Review Management
   5. Reports & Analytics
   0. Exit

   Enter your choice: 1

   --- Product Management ---
   1. Search Products
   2. Filter by Category
   3. Add New Product
   4. Update Product
   5. View Low Stock Items
   0. Back to Main Menu
   Select an option: 1
   Enter product name to search: iPhone 15 Pro

   Search Results:
   ID: 101, Name: iPhone 15 Pro, Price: $999.99, Stock: 50, Brand: Apple

   --- Product Management ---
   1. Search Products
   2. Filter by Category
   3. Add New Product
   4. Update Product

5. View Low Stock Items
0. Back to Main Menu
Select an option: 2
Enter category ID to filter by: 1

Products in Category:
ID: 101, Name: iPhone 15 Pro, Price: $999.99, Stock: 50, Category: Electronics, Brand: Apple
ID: 102, Name: Samsung Galaxy S24, Price: $899.99, Stock: 30, Category: Electronics, Brand: Samsung
ID: 103, Name: MacBook Air M2, Price: $1299.99, Stock: 25, Category: Electronics, Brand: Apple

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option: 3
Product name: dog
Category ID: 1
Price: 1
Stock quantity: 1
Description: dog
Brand: dog
Weight (kg): 1
Product added with ID 112.

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option: 4

```
Enter product ID to update: 112
Leave a field blank to keep current value.
New price: 2
New stock quantity:
New description:
New brand:
New weight (kg):
Product updated.

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option:
Invalid option. Please try again.

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option: 5
Enter stock threshold (default 50): 10

Low Stock Products:
ID: 111, Name: tyler, Stock: 1, Brand: awesome
ID: 112, Name: dog, Stock: 1, Brand: dog

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
```

5. View Low Stock Items
0. Back to Main Menu
Select an option: 0
E-COMMERCE MANAGEMENT SYSTEM
1. Product Management
2. Customer Management
3. Order Processing
4. Review Management
5. Reports & Analytics
0. Exit

Enter your choice: 2

--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 1
First name: me
Last name: me
Email: me
Phone: 1233211234
Date of birth (YYYY-MM-DD): 2001-09-12
Customer registered with ID 2.

--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 2
Enter customer ID: 2
Customer ID: 2
Name: me me

```
Email: me
Phone: 1233211234
DOB: 2001-09-12
Registered: 2025-12-03
Active: True

--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 3
Enter customer ID to update: 2
Leave field blank to skip.
New phone:
New email:
No updates provided.

--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 4
Enter customer ID: 2
No orders found for this customer.

--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 0
E-COMMERCE MANAGEMENT SYSTEM
1. Product Management
```

2. Customer Management
3. Order Processing
4. Review Management
5. Reports & Analytics
0. Exit

Enter your choice: 3

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 1
Customer ID: 2
Shipping address: 222
Payment method: card
Order created with ID 2.

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 3
Order ID: 2
New status (Pending/Shipped/Delivered/Cancelled): shipped
Order status updated.

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status

4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 4
Order ID to cancel: 2
Order not found or has no items to cancel.
Failed to cancel order.

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 2
Order ID: 2
Product ID: 111
Quantity: 2
Insufficient stock. Only 1 units available.
Failed to add item.

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 2
Order ID: 111
Product ID: 1
Quantity: 1
Product not found or inactive.
Failed to add item.

--- Order Processing ---
1. Create New Order

2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 1
Customer ID:
Invalid customer ID.

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 5
Order ID: 2

Order Details:
Order ID: 2, Date: 2025-12-03, Total: $0.00, Status: shipped
Customer: me me (ID 2)

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 0
E-COMMERCE MANAGEMENT SYSTEM
1. Product Management
2. Customer Management
3. Order Processing
4. Review Management
5. Reports & Analytics

0. Exit

Enter your choice: 4

--- Review Management ---
1. Add Review
2. View Product Reviews
3. Get Average Rating
4. Verify Review (Admin)
0. Back to Main Menu
Select an option: 1
Customer ID: 2
Product ID: 111
Rating (1-5): 4
Review text: good
Review added with ID 3.

--- Review Management ---
1. Add Review
2. View Product Reviews
3. Get Average Rating
4. Verify Review (Admin)
0. Back to Main Menu
Select an option: 2
Enter product ID: 111

Product Reviews:
ID: 3, Rating: 4, Reviewer: me me, Date: 2025-12-03, Status: Unverified
  Review: good

--- Review Management ---
1. Add Review
2. View Product Reviews
3. Get Average Rating
4. Verify Review (Admin)
0. Back to Main Menu
Select an option: 3

```
Enter product ID: 111
Average Rating: 4.00 based on 1 review(s).

--- Review Management ---
1. Add Review
2. View Product Reviews
3. Get Average Rating
4. Verify Review (Admin)
0. Back to Main Menu
Select an option: 4
Enter review ID to verify: 2
Review verified.

--- Review Management ---
1. Add Review
2. View Product Reviews
3. Get Average Rating
4. Verify Review (Admin)
0. Back to Main Menu
Select an option: 0
E-COMMERCE MANAGEMENT SYSTEM
1. Product Management
2. Customer Management
3. Order Processing
4. Review Management
5. Reports & Analytics
0. Exit


Enter your choice: 5
Invalid choice. Please try again.
E-COMMERCE MANAGEMENT SYSTEM
1. Product Management
2. Customer Management
3. Order Processing
4. Review Management
5. Reports & Analytics
0. Exit
```

Enter your choice: 0
Thank you for using E-Commerce System!
Database connection closed.

Process finished with exit code 0

```
C:\Users\brett\PycharmProjects\EcommerceApp\venv\Scripts\python.exe C:\Users\brett\PycharmProjects\EcommerceApp\main.py
Starting E-Commerce Application...
Database connection successful!
E-COMMERCE MANAGEMENT SYSTEM
1. Product Management
2. Customer Management
3. Order Processing
4. Review Management
5. Reports & Analytics
0. Exit


Enter your choice: 1

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option: 1
Enter product name to search: iPhone 15 Pro
```

```
Search Results:
ID: 101, Name: iPhone 15 Pro, Price: $999.99, Stock: 50, Brand: Apple

--- Product Management ---
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3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option: 2
Enter category ID to filter by: 1

Products in Category:
ID: 101, Name: iPhone 15 Pro, Price: $999.99, Stock: 50, Category: Electronics, Brand: Apple
ID: 102, Name: Samsung Galaxy S24, Price: $899.99, Stock: 30, Category: Electronics, Brand: Samsung
ID: 103, Name: MacBook Air M2, Price: $1299.99, Stock: 25, Category: Electronics, Brand: Apple
```

```
--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option: 3
Product name: dog
Category ID: 1
Price: 1
Stock quantity: 1
Description: dog
Brand: dog
Weight (kg): 1
Product added with ID 112.

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option: 4
Enter product ID to update: 112
Leave a field blank to keep current value.
New price: 2
New stock quantity:
```

```
New description:
New brand:
New weight (kg):
Product updated.

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option:
Invalid option. Please try again.

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option: 5
Enter stock threshold (default 50): 10
```

```
Low Stock Products:
ID: 111, Name: tyler, Stock: 1, Brand: awesome
ID: 112, Name: dog, Stock: 1, Brand: dog

--- Product Management ---
1. Search Products
2. Filter by Category
3. Add New Product
4. Update Product
5. View Low Stock Items
0. Back to Main Menu
Select an option: 0
E-COMMERCE MANAGEMENT SYSTEM
1. Product Management
2. Customer Management
3. Order Processing
4. Review Management
5. Reports & Analytics
0. Exit


Enter your choice: 2

--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 1
First name: me
```

```
Last name: me
Email: me
Phone: 1233211234
Date of birth (YYYY-MM-DD): 2001-09-12
Customer registered with ID 2.

--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 2
Enter customer ID: 2
Customer ID: 2
Name: me me
Email: me
Phone: 1233211234
DOB: 2001-09-12
Registered: 2025-12-03
Active: True
```

```
--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 3
Enter customer ID to update: 2
Leave field blank to skip.
New phone:
New email:
No updates provided.

--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 4
Enter customer ID: 2
No orders found for this customer.

--- Customer Management ---
1. Register New Customer
2. View Customer Profile
3. Update Customer Information
4. View Customer Orders
0. Back to Main Menu
Select an option: 0
```

```
E-COMMERCE MANAGEMENT SYSTEM
1. Product Management
2. Customer Management
3. Order Processing
4. Review Management
5. Reports & Analytics
0. Exit


Enter your choice: 3

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 1
Customer ID: 2
Shipping address: 222
Payment method: card
Order created with ID 2.
```

```
--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 3
Order ID: 2
New status (Pending/Shipped/Delivered/Cancelled): shipped
Order status updated.

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 4
Order ID to cancel: 2
Order not found or has no items to cancel.
Failed to cancel order.
```

```
--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 2
Order ID: 2
Product ID: 111
Quantity: 2
Insufficient stock. Only 1 units available.
Failed to add item.

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 2
Order ID: 111
Product ID: 1
Quantity: 1
Product not found or inactive.
Failed to add item.
```

```
--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 2
Order ID: 111
Product ID: 1
Quantity: 1
Product not found or inactive.
Failed to add item.

--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 1
Customer ID:
Invalid customer ID.
```

```
--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 5
Order ID: 2

Order Details:
Order ID: 2, Date: 2025-12-03, Total: $0.00, Status: shipped
Customer: me me (ID 2)
```

```
--- Order Processing ---
1. Create New Order
2. Add Item to Order
3. Update Order Status
4. Cancel Order
5. View Order Details
0. Back to Main Menu
Select an option: 0
E-COMMERCE MANAGEMENT SYSTEM
1. Product Management
2. Customer Management
3. Order Processing
4. Review Management
5. Reports & Analytics
0. Exit


Enter your choice: 4
```

```
--- Review Management ---
1. Add Review
2. View Product Reviews
3. Get Average Rating
4. Verify Review (Admin)
0. Back to Main Menu
Select an option: 1
Customer ID: 2
Product ID: 111
Rating (1-5): 4
Review text: good
Review added with ID 3.

--- Review Management ---
1. Add Review
2. View Product Reviews
3. Get Average Rating
4. Verify Review (Admin)
0. Back to Main Menu
Select an option: 2
Enter product ID: 111
```

```
Product Reviews:
ID: 3, Rating: 4, Reviewer: me me, Date: 2025-12-03, Status: Unverified
  Review: good

--- Review Management ---
1. Add Review
2. View Product Reviews
3. Get Average Rating
4. Verify Review (Admin)
0. Back to Main Menu
Select an option: 3
Enter product ID: 111
Average Rating: 4.00 based on 1 review(s).

--- Review Management ---
1. Add Review
2. View Product Reviews
3. Get Average Rating
4. Verify Review (Admin)
0. Back to Main Menu
Select an option: 4
Enter review ID to verify: 2
Review verified.
```

ect. – the output is the same as the text I got - I just wanted to show the output from the IDE to show I didn't make it up or something